



## SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences  
Chennai- 602105



Student Name: Vimala K

Reg. No:192424276

Course Code: DSA0306

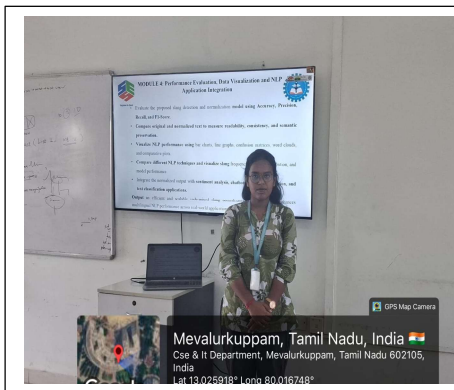
Slot: C

Course Name: NATURAL LANGUAGE PROCESSING

Course Faculty: Dr. T. Kumaragurubaran

**Project Title: Code-Mixed Slang Detection and Normalization Using Language Identification, Named Entity Recognition, and Transformer-Based NLP**

### Module Photographs:





#### MODULE 4: Performance Evaluation, Data Visualization and NLP Application Integration

Engineer to Excel

- Evaluate the proposed slang detection and normalization model using Accuracy, Precision, Recall, and F1-Score.
- Compare original and normalized text to measure readability, consistency, and semantic preservation.
- Visualize NLP performance using bar charts, line graphs, confusion matrices, word clouds, and comparative plots.
- Compare different NLP techniques and visualize slang frequency, language distribution, and model performance.
- Integrate the normalized output with sentiment analysis, chatbots, machine translation, and text classification applications.

Output an efficient and scalable code-mixed slang normalization framework that enhances multilingual NLP performance across real-world applications.

| CONTEXT INTELLIGENCE |                        |                    |
|----------------------|------------------------|--------------------|
| TARGET               | CONTEXT                | INTERPRETATION     |
| dei                  | .. bro semma ..        | Informal Noun/Verb |
| bro                  | .. dei semma movie ..  | Informal Adjective |
| seemma               | .. dei bro movie da .. | Informal Adjective |
| da                   | .. semma movie ..      | Informal Adjective |

### Project Description:

The project, **Code-Mixed Slang Detection and Normalization Using Language Identification, Named Entity Recognition, and Transformer-Based NLP**, focuses on improving the understanding of informal and multilingual text commonly used on social media and messaging platforms. Code-mixed text contains words from two or more languages, making it difficult for traditional NLP systems to process accurately. The system first performs text preprocessing and language identification to detect the languages used in the text. Named Entity Recognition is then applied to identify important entities such as person names, locations, and organizations. Slang detection techniques are used to identify informal, abbreviated, or non-standard words. Finally, Transformer-based NLP models normalize these slang words into meaningful standard language while preserving the original context and meaning. The proposed system can improve applications such as chatbots, sentiment analysis, machine translation, text classification, and social media analysis.

Student Signature

Guide Signature